funTDA — Part C: structural images (lateral ventricles, Parkinson's disease)

Distance-transform filtrations, H_1 and H_2 landscapes, FPCA and exhaustive permutation tests over 16 PPMI subjects

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Table of contents

What this document is

Parts A and B of this repository apply the funTDA pipeline to networks and to simulated time series, in both cases reading H_1. Part C extends it in two directions at once: the input encoding becomes a cubical complex built from a three-dimensional image, and the homological order rises to $r = 2$, so that the object being measured is a genuine anatomical cavity rather than a cycle.

The question is whether the lateral ventricles differ between Parkinson's disease and control, and whether H_2 adds discriminative power over H_1.

Every number below is recomputed in R from the committed CSVs in `output/`. Where a value was also produced by the Python pipeline, `stopifnot()` compares the two implementations.

Data

Table 1: The cohort. Ventricular volume is the main definition, labels 4 and 43.

subject	group	age	ventricle (mL)	components	eTIV (mL)
sub-01	PD	59	37.1	2	1571.5
sub-02	PD	58	21.6	2	1657.3
sub-03	PD	70	28.3	3	1407.1
sub-04	Control	62	15.1	3	1504.8
sub-05	PD	61	16.6	3	1418.4
sub-06	Control	54	9.3	5	1385.8
sub-07	PD	76	44.9	2	1623.9
sub-08	Control	46	13.0	3	1656.3
sub-09	PD	56	17.1	1	1452.6
sub-10	Control	58	19.3	3	1506.4
sub-11	Control	74	50.2	1	1551.8
sub-12	PD	66	8.3	2	1415.9
sub-13	Control	69	37.9	2	1446.3
sub-14	PD	63	15.3	2	1499.1
sub-15	Control	69	22.5	2	1353.0
sub-16	Control	59	16.7	2	1431.1

Sixteen PPMI participants, eight control and eight with Parkinson's disease, all at the baseline visit, all scanned with the same SAG 3D MPRAGE protocol at 1.0 mm isotropic resolution. Nine are men and seven women.

Table 2: Sex by group. Fisher exact p = 1.000.

group	M	F
Control	5	3
PD	4	4

Sex is not carried per subject in this repository. The fact that matters for the comparison is that it is balanced between groups, and that is what the table above reports; a per-subject sex column would sharpen the quasi-identifier without being read by any analysis.

The DICOM themselves are not in this repository and cannot be: they are identified human data under a Data Use Agreement that does not permit redistribution. What is committed is the topological summary — the birth-death pairs and the landscapes — which is two steps removed from the image and carries no identifier. Subject numbers are replaced by `sub-01` … `sub-16`; `images/export_for_repo.py` performs the substitution and writes the mapping to a file that stays on the local machine.

Group and age are kept, because age is needed for the positive control below and because sixteen (group, age) pairs drawn from a cohort of several thousand are what any published table carries. Sex is not kept per subject: it is reported in aggregate above, no analysis reads it, and on this cohort (group, age) already makes fourteen of the sixteen rows unique, so carrying sex as well would add re-identification surface for no analytical return.

```
Age  Control 61.38 +- 9.13   PD 63.62 +- 6.74   difference +2.25 years, Welch p = 0.585
```

```
Ventricular volume  Control 23.01 mL   PD 23.64 mL   difference +0.63 mL
```

The two groups are balanced on age: the difference is 2.25 years and does not approach significance. Age is therefore not a confounder in the classical sense here. What it is, as the positive control below shows, is a source of within-group variance.

What the filtration measures

Tissue is defined as the complement of the cavity, and the filtration is by sublevel sets of

$$d(x) = \mathrm{dist}(x,\, C^c),$$

computed inside the segmented ventricle with physical sampling equal to the voxel size. Two facts follow, and together they are what makes the persistence readable in millimetres.

Since $\{d \leq 0\} = C^c$, every H_2 class associated with a bounded component of $\mathrm{int}(C)$ is **born at** $t = 0$. And the largest death is $\max_{x \in C} d(x)$, which is the **radius of the largest inscribed ball**. The maximum persistence of H_2 is therefore that radius, exactly. Neither statement uses convexity, and both carry over to cubical complexes with d the discrete Euclidean distance transform.

This matters because it is not true of the more obvious construction. If tissue is taken to be the brain mask minus the cavity, the class is born late whenever the segmentation touches the edge of that mask, and the measured value falls short of the inradius — by up to 2.24 mm on this cohort, against a between-group difference of 0.11 mm.

Figure 1 is what the pipeline produces. The red region is FreeSurfer labels 4 and 43, and the third column is that region on its own: a closed surface bounding a cavity, which is the H_2 class the rest of this document measures. The two subjects differ by twenty years and by a factor of five in volume, and both are controls.

```
Inradius, native (eTIV-corrected)  Control 6.308 +- 1.627 mm   PD 6.414 +- 1.431 mm   differen

Inradius, MNI 1 mm                 Control 6.823 +- 1.756 mm   PD 7.025 +- 1.611 mm   differen
```

```
Correlation between age and inradius: r = 0.750
```

The difference between clinical groups is a tenth of a millimetre against standard deviations above one and a half. The correlation with age is 0.750.

From diagrams to landscapes

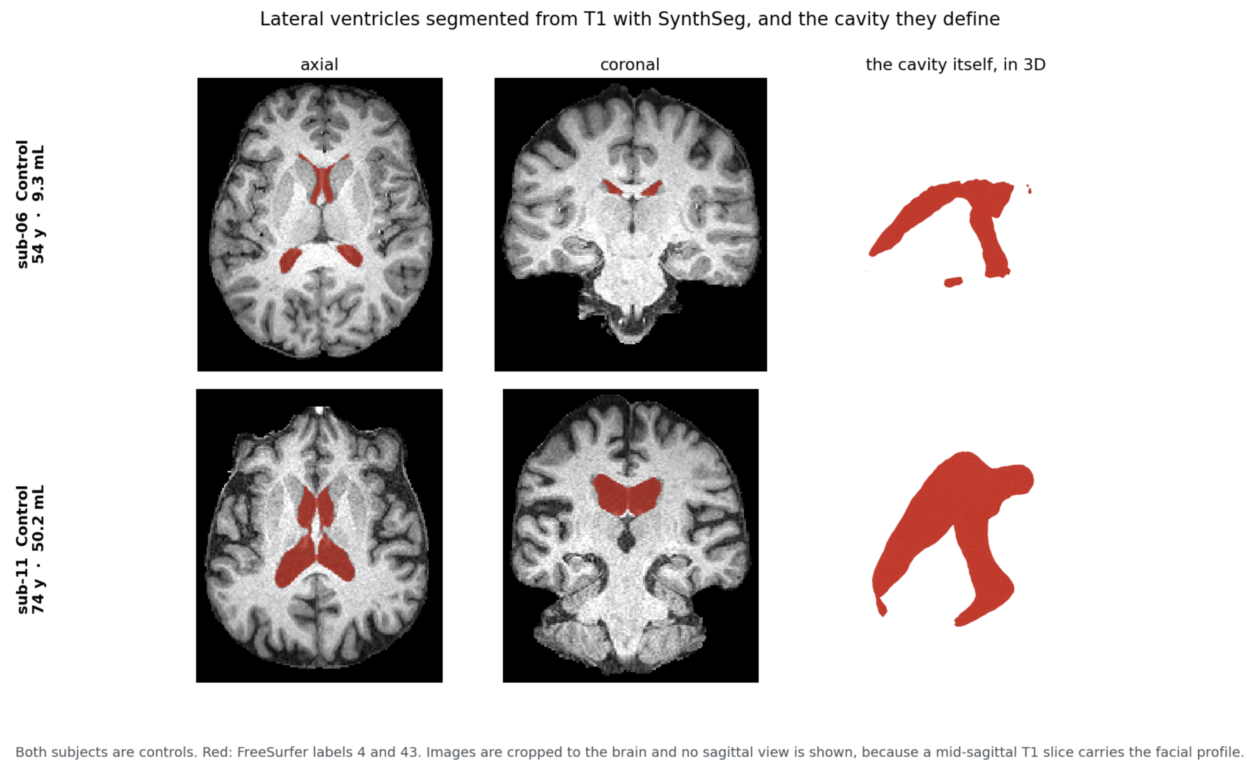

Figure 1: Lateral ventricles segmented from T1 with SynthSeg, and the cavity they define. Both subjects are controls, aged 54 and 74.

combination	max abs diff	worst subject	its diff
native_main_1	9.714e-16	sub-01	9.714e-16
native_main_2	2.220e-15	sub-03	2.220e-15
native_full_1	9.714e-16	sub-01	9.714e-16
native_full_2	2.220e-15	sub-03	2.220e-15
mni_main_1	2.678e-15	sub-13	2.678e-15
mni_main_2	2.665e-15	sub-13	2.665e-15
mni_full_1	2.678e-15	sub-13	2.678e-15
mni_full_2	2.665e-15	sub-13	2.665e-15

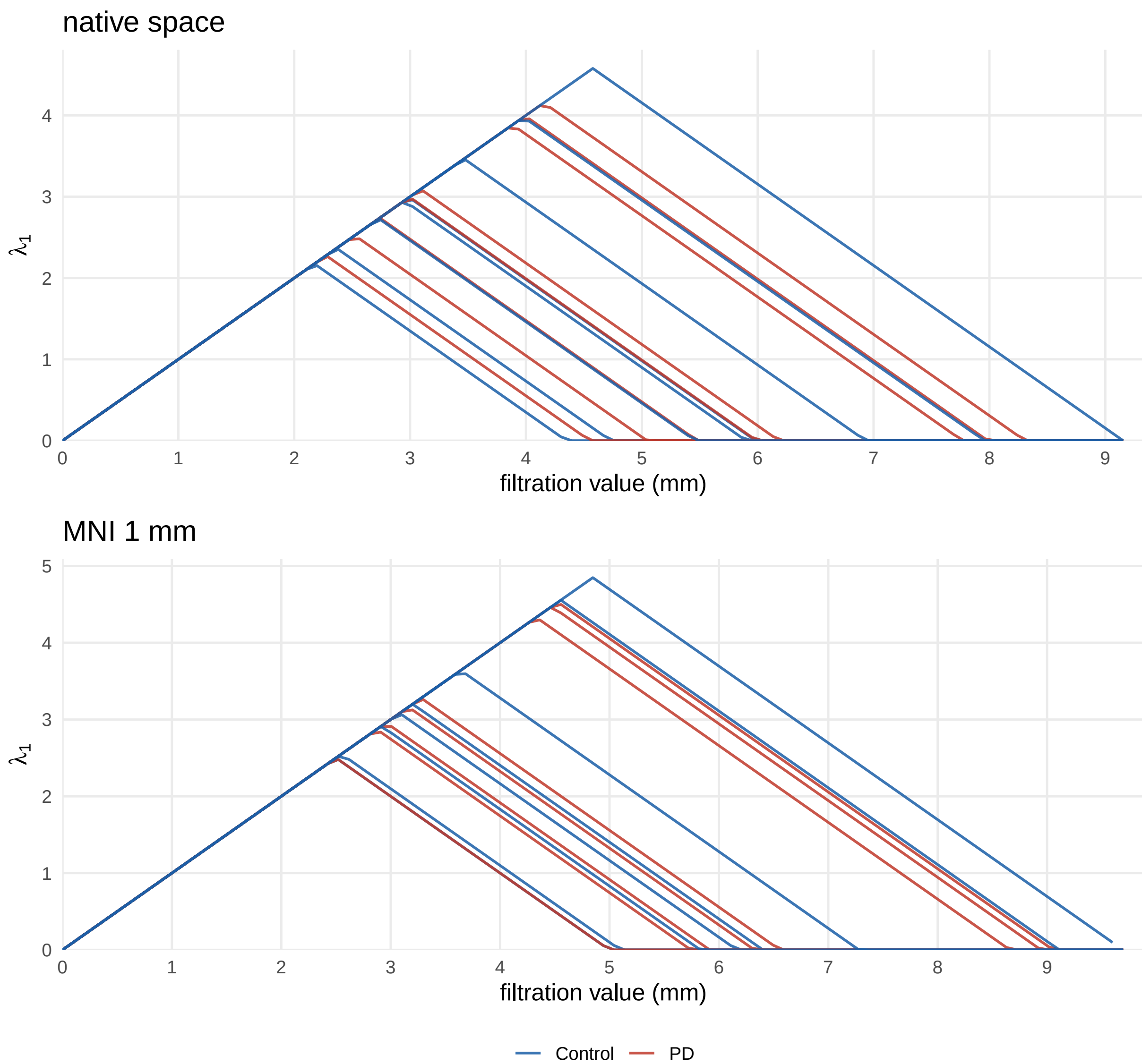

Figure 2: First landscape layer for H_2, one curve per subject, main cavity definition. The filtration value is a length: the peak of a tent sits at half the inradius of the cavity that produced it.

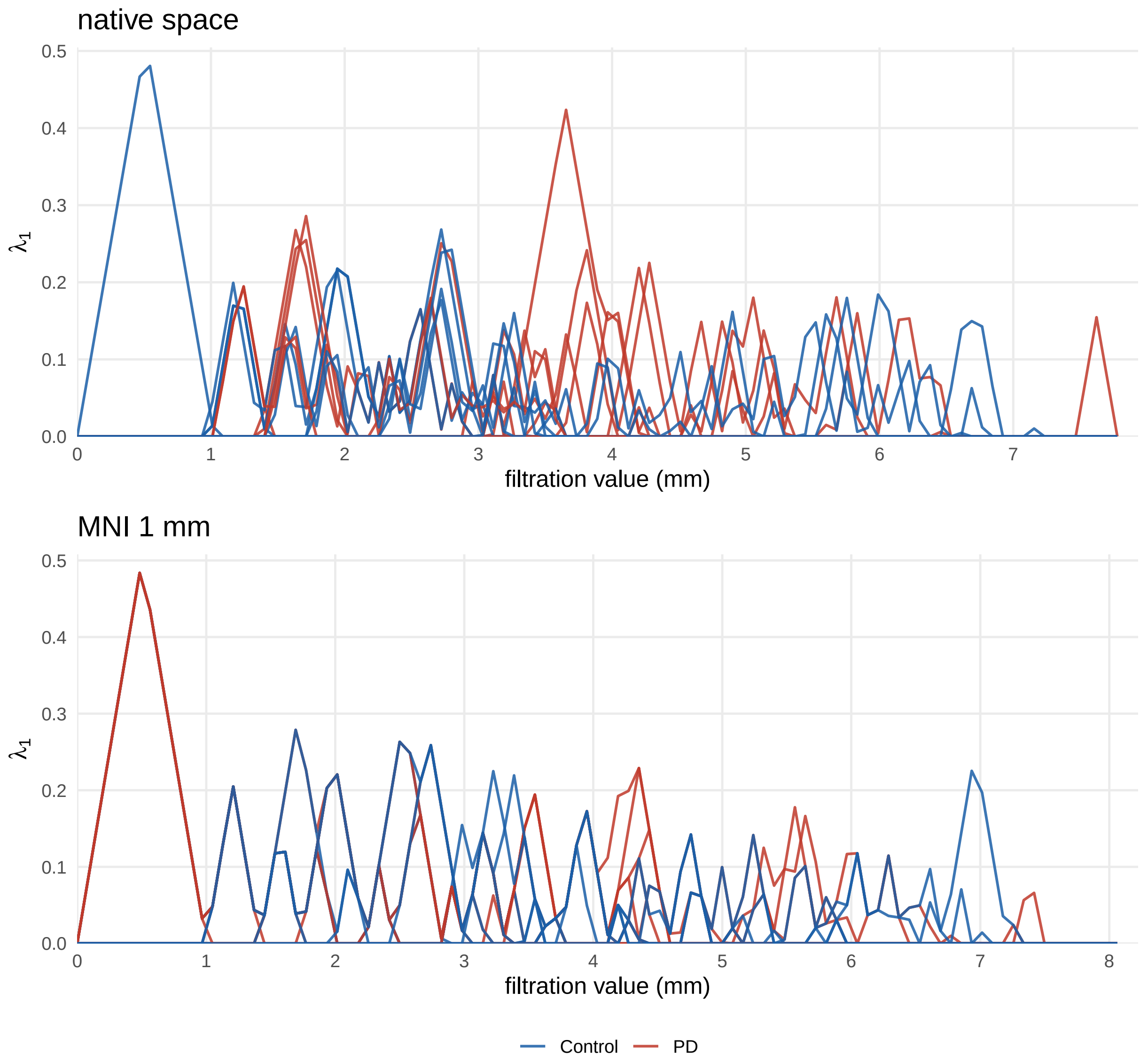

Figure 3: The same for H_1. The cycles carry a good deal more structure than the cavity, and rather less of it separates the two groups.

Group means are deliberately not drawn. With eight curves per group and this much spread, a mean with a standard-error band suggests a separation that the individual curves do not show.

Functional principal component analysis

```
Native, main, H2:  PC1 79.4%   PC2 7.8%   cumulative 87.2%
```

The quadrature weight $\sqrt{\Delta x}$ is applied before the decomposition so that the scores are the L^2 inner products of Proposition `prop:kl-expansion` rather than Euclidean inner products of the discretised vectors. On a uniform grid this leaves the explained-variance ratios unchanged but puts the eigenvalues ν_p in the right units.

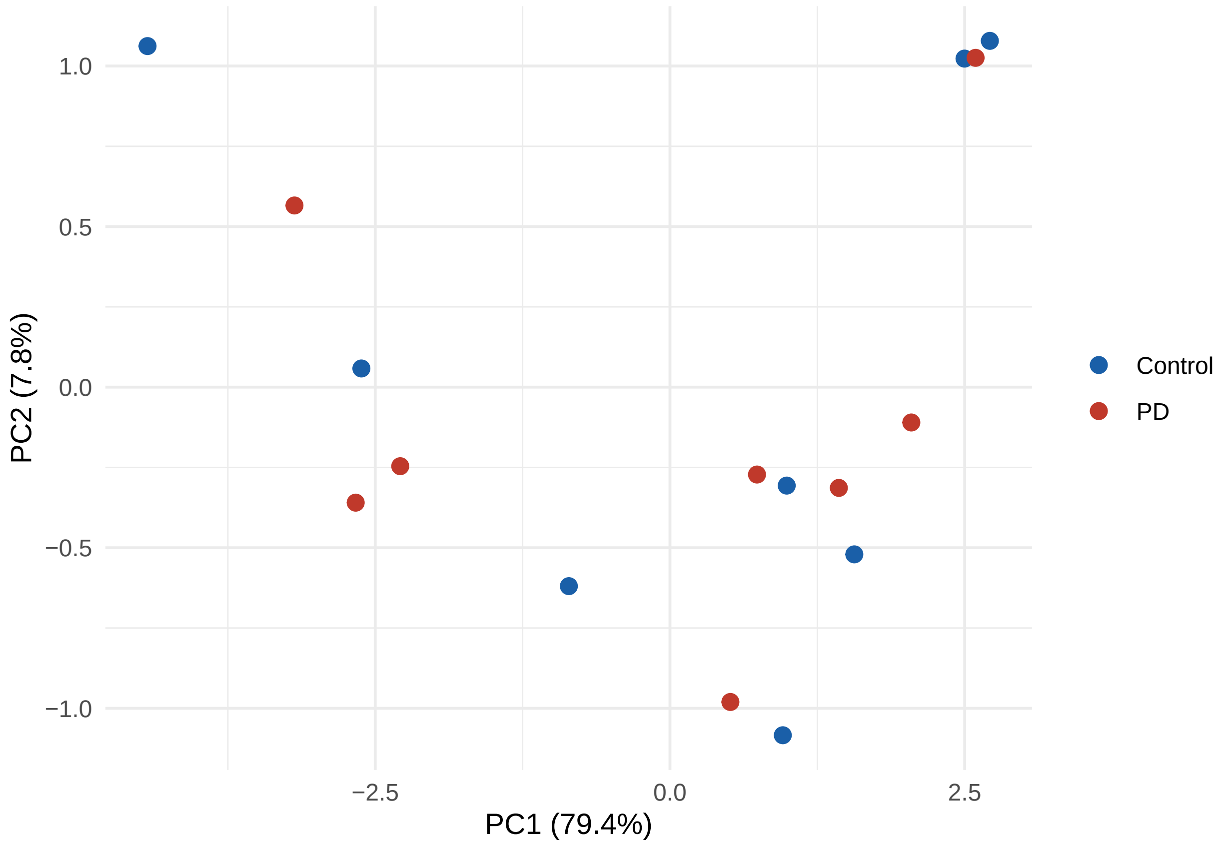

Figure 4: FPC scores, native space, H_2, main definition. No grouping structure is visible.

The pre-registered test

The statistic is the supremum, over the five landscape layers and over the grid, of the pointwise Welch statistic of Definition `def:functional-ttest`. It is calibrated by the **exhaustive** set of $\binom{16}{8} = 12{,}870$ label assignments rather than by a sample of them, so the null distribution is the exact one for this design.

Table 3: Pre-registered supremum statistic, and the post-hoc L^p distances.

space	definition	homology	T	crit95	p	L1	L2	Linf
native	main	1	2.3344	3.7866	0.6134	0.1809	0.1299	0.2654
native	main	2	1.9458	3.7707	0.9251	0.9459	0.9523	0.9447
native	full	1	2.3344	4.0432	0.7307	0.2496	0.3304	0.8127
native	full	2	3.2890	3.9584	0.1716	0.9074	0.8810	0.7260
mni	main	1	2.1995	4.5826	0.8219	0.5981	0.6469	0.7716
mni	main	2	2.0494	4.2426	0.9584	0.9324	0.9389	0.9546
mni	full	1	2.1995	4.5826	0.8670	0.3605	0.2842	0.5806

space	definition	homology	T	crit95	p	L1	L2	Linf
mni	full	2	2.8761	4.4840	0.4424	0.8881	0.8906	0.7828

combination	T (R)	T (Python)	dT	p (R)	p (Python)	dp
native_main_2	1.9458	1.9458	-0.0000	0.9251	0.9251	-0.0000
native_main_1	2.3344	2.3344	-0.0000	0.6134	0.6134	-0.0000
mni_main_2	2.0494	2.0494	-0.0000	0.9584	0.9584	-0.0000
mni_main_1	2.1995	2.1995	-0.0000	0.8219	0.8219	+0.0000
native_full_2	3.2890	3.2890	+0.0000	0.1716	0.1716	-0.0000
mni_full_2	2.8761	2.8761	+0.0000	0.4424	0.4424	+0.0000

Largest discrepancy between the two implementations: 4.72e-05

The two implementations agree to within 5e-3 on every checked value.

Smallest reachable p-value with this design: 0.000155 Smallest p-value observed across the eight contrasts: 0.1716

Nothing rejects. The primary contrast, H_2 in the main cavity definition, gives $p = 0.925$ in native space and $p = 0.958$ in MNI. The two spaces and the two cavity definitions were both pre-registered to be reported whatever they showed, and they agree.

Age as a positive control — declared post hoc

A null says nothing unless the machinery can detect a signal known to be present. Age correlates with the inradius at 0.750 in this cohort, so a median split on age is a signal that must be there. It also happens to be balanced by diagnosis, four control and four patients on each side, so the age contrast is not itself the group contrast in disguise.

Table 4: Age split. Compare with the diagnosis columns above.

space	definition	homology	sup_t	L1	L2	Linf
native	main	1	0.3231	0.0651	0.1417	0.2726
native	main	2	0.0755	0.0466	0.0485	0.0580
native	full	1	0.2286	0.3166	0.4973	0.5086
native	full	2	0.1064	0.0387	0.0476	0.0580
mni	main	1	0.5946	0.4695	0.5882	0.7916
mni	main	2	0.0749	0.0581	0.0606	0.0721
mni	full	1	0.6513	0.6387	0.7355	0.7832
mni	full	2	0.0985	0.0488	0.0587	0.0721

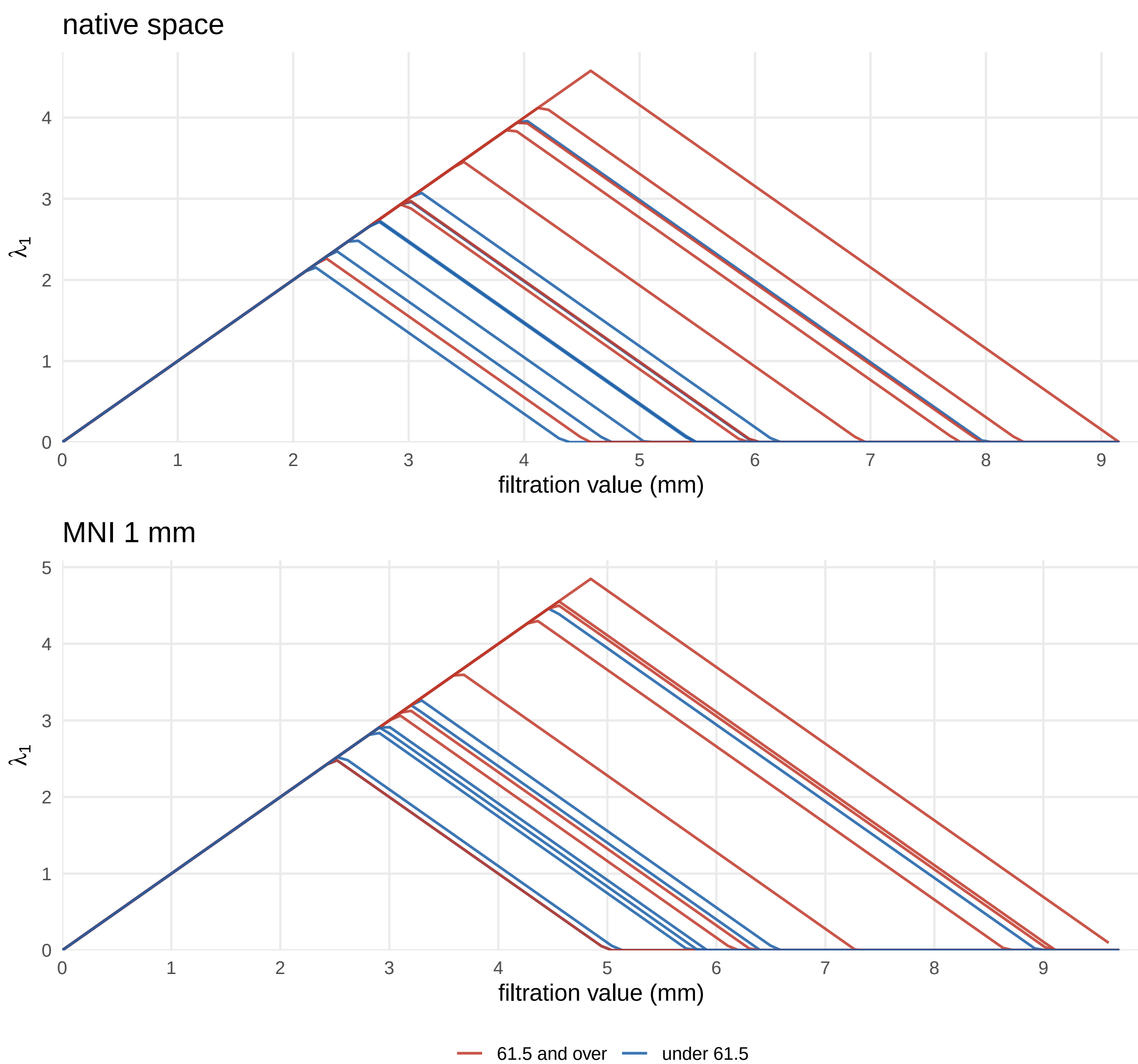

Figure 5: The same H_2 curves as above, coloured by a median split on age instead of by diagnosis. This is the contrast the statistic responds to.

With the parameter-free L^p statistics, H_2 places age at the 5 % boundary in all four space-by-definition combinations and under all three norms, while diagnosis does not fall below 0.72 anywhere. That three different norms agree is the argument that no tuning is doing the work.

H_1 does not show the age effect. The cavity picks up the change that is known to be there; the cycles do not.

The two contrasts share a null. It is the set of L^2 distances obtained by relabelling the same sixteen subjects in all $\binom{16}{8} = 12{,}870$ ways, and it does not know which labelling is the real one; only the observed statistic does. Drawing both observed values against that single null puts the positive control and the clinical result on one axis.

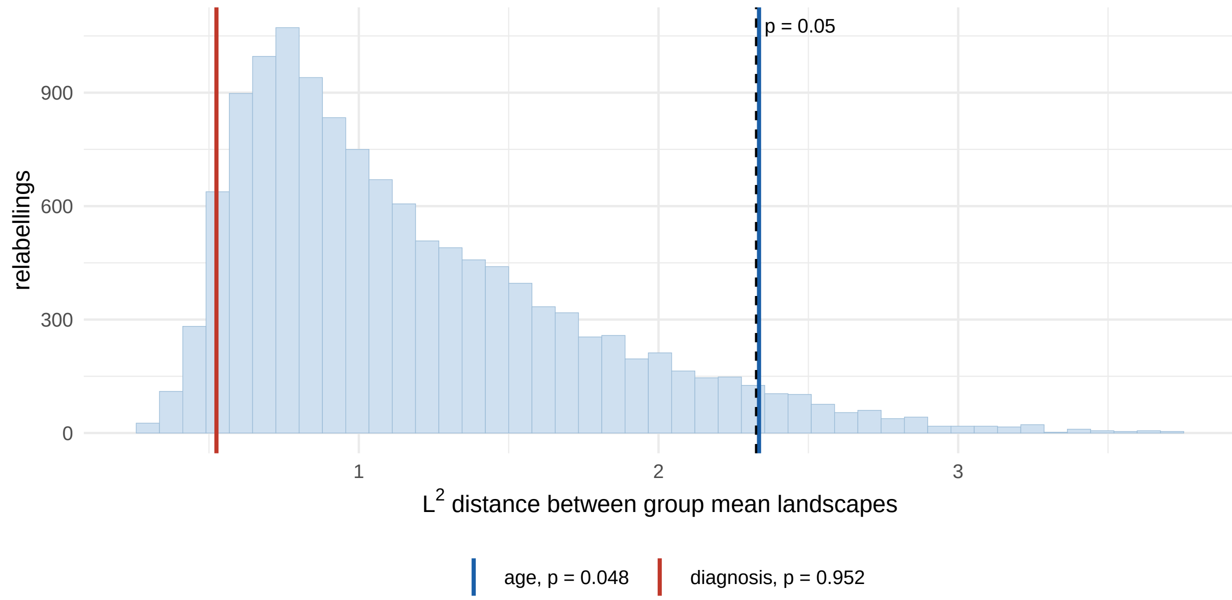

Figure 6: Exact permutation null of the POST HOC L^2 statistic, not of the pre-registered supremum: native frame, main cavity definition, H_2. All 12,870 relabellings of the same sixteen subjects, with the age and diagnosis statistics marked. Dashed line: the 95th percentile of the null, above which a statistic attains $p \leq 0.05$.

Where the supremum falls

Table 5: How many of the sixteen curves are non-zero at the point where the supremum is attained.

key	layer	nonzero	of	active_cols	total_cols
native_main_1	1	5	16	386	505
native_main_2	5	3	16	399	505
native_full_1	1	5	16	398	505
native_full_2	3	16	16	403	505
mni_main_1	1	10	16	390	505
mni_main_2	4	3	16	413	505
mni_full_1	1	10	16	390	505
mni_full_2	4	14	16	413	505

The supremum of a pointwise **standardised** statistic over functions of compact support is unstable at the edge of that support: almost every curve is exactly zero there, the pooled variance collapses, and the ratio is driven by the two or three that are not. On the main definition the supremum lands where only three of the sixteen curves are non-zero.

Restricting the supremum to a well-supported region does fix this, but the threshold has no canonical value and the p-value moves with it. The pre-registered statistic is therefore reported unrestricted, and the L^p distances are reported beside it as the alternative that carries no free parameter. They agree on every contrast.

Provenance for the remaining figures quoted in Chapter 5

Everything the chapter states that is not already produced above is produced here, so that no number in it rests on a calculation done once by hand.

```
Components added by the full definition: 31, inradii 0.97 to 2.82 mm, median 1.75

Bars born at t = 0, native main: 38 of 398; equal to the component count in every subject: TRU

Component counts agree between native and MNI: TRUE

Largest finite death over the eight combinations: 9.6954 mm

Calibration of the tie count (the plateau across tolerances is the number):

  native_main_2    T = 1.945762   counts at 1e-12/1e-9/1e-6/1e-4: 126 / 126 / 126 / 126
                   ten smallest |null - T|: 4.44e-16 4.44e-16 4.44e-16 4.44e-16 4.44e-16 4.44e

  mni_main_2       T = 2.049390   counts at 1e-12/1e-9/1e-6/1e-4: 296 / 296 / 296 / 296
                   ten smallest |null - T|: 0.00e+00 0.00e+00 0.00e+00 0.00e+00 0.00e+00 0.00e

Ties: native 126 (p 0.925, 0.915 discarding them), MNI 296 (p 0.958, 0.935)

Correlation of each layer peak with the first: +1.00 -0.16 -0.68 -0.67 -0.44

Group effect with age and group fitted jointly: -0.218 mm

eTIV Control 1479.4 mL vs PD 1505.7 mL, Welch p = 0.602

n = 14: eight contrasts, lowest p = 0.42; inradius difference +0.33 mm against +0.11 with sixt

Largest B-spline representation error over the eight combinations: 1.2646e-02 mm
```

Sanity checks

- PASS landscape layers are non-negative
- PASS layers are ordered, lambda_1 >= lambda_2 >= … >= lambda_K
- PASS the second-order basis represents the landscape to within 1.26e-02 mm, over all eight combinations and all sixteen subjects
- PASS the basis has exactly L = Q - 1 functions
- PASS every H2 class in the main definition is born at zero or later
- PASS maximum H2 persistence equals maximum death when births are zero
- PASS the permutation set is exhaustive and has the expected size
- PASS each statistic value appears an even number of times in the null
- PASS no p-value falls below the smallest reachable value
- PASS the degenerate-column guard is insensitive to its tolerance
- PASS the statistic is stable under a machine-epsilon perturbation

The penultimate check is worth a word. With two groups of equal size, a selection and its complement produce the same statistic, so every value in the null distribution occurs an even number of times. That is why the smallest reachable p-value is 2/12870 = 0.000155 and not 1/12870.

What this part shows

The chain runs end to end and is reproducible from the committed summaries. The maximum persistence of H_2 under this filtration is the inradius of the cavity, exactly, which gives the topological summary a unit and a meaning.

With eight subjects per group no difference is detected between Parkinson's disease and control, by H_1, by H_2, or by ventricular volume, in either space and under either cavity definition. The positive control places the detection floor of the design at roughly 1.4 mm of inradius; the observed between-group difference is 0.11 mm.

```r
sessionInfo()
```

```
R version 4.5.3 (2026-03-11)
Platform: aarch64-apple-darwin20
Running under: macOS Tahoe 26.5.2

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LA

locale:
[1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8

time zone: Europe/Madrid
tzcode source: internal

attached base packages:
```

```
[1] splines   stats     graphics  grDevices utils     datasets  methods
[8] base

other attached packages:
[1] patchwork_1.3.2 ggplot2_4.0.3   fda_6.3.0       deSolve_1.42
[5] fds_1.9         RCurl_1.98-1.19 rainbow_3.8     pcaPP_2.0-5
[9] MASS_7.3-65

loaded via a namespace (and not attached):
 [1] Matrix_1.7-4       gtable_0.3.6       jsonlite_2.0.0     dplyr_1.2.1
 [5] compiler_4.5.3     tidyselect_1.2.1   ks_1.15.2          bitops_1.0-9
 [9] cluster_2.1.8.2    scales_1.4.0       yaml_2.3.12        fastmap_1.2.0
[13] lattice_0.22-9     R6_2.6.1           labeling_0.4.3     generics_0.1.4
[17] knitr_1.51         tibble_3.3.1       pillar_1.11.1      RColorBrewer_1.1-3
[21] rlang_1.2.0        hdrcde_3.5.0       xfun_0.57          S7_0.2.2
[25] cli_3.6.6          withr_3.0.2        magrittr_2.0.5     digest_0.6.39
[29] grid_4.5.3         mvtnorm_1.4-2      mclust_6.1.3       lifecycle_1.0.5
[33] vctrs_0.7.3        KernSmooth_2.23-26 evaluate_1.0.5     pracma_2.4.6
[37] glue_1.8.1         farver_2.1.2       codetools_0.2-20   colorspace_2.1-3
[41] rmarkdown_2.31     pkgconfig_2.0.3    tools_4.5.3        htmltools_0.5.9
```